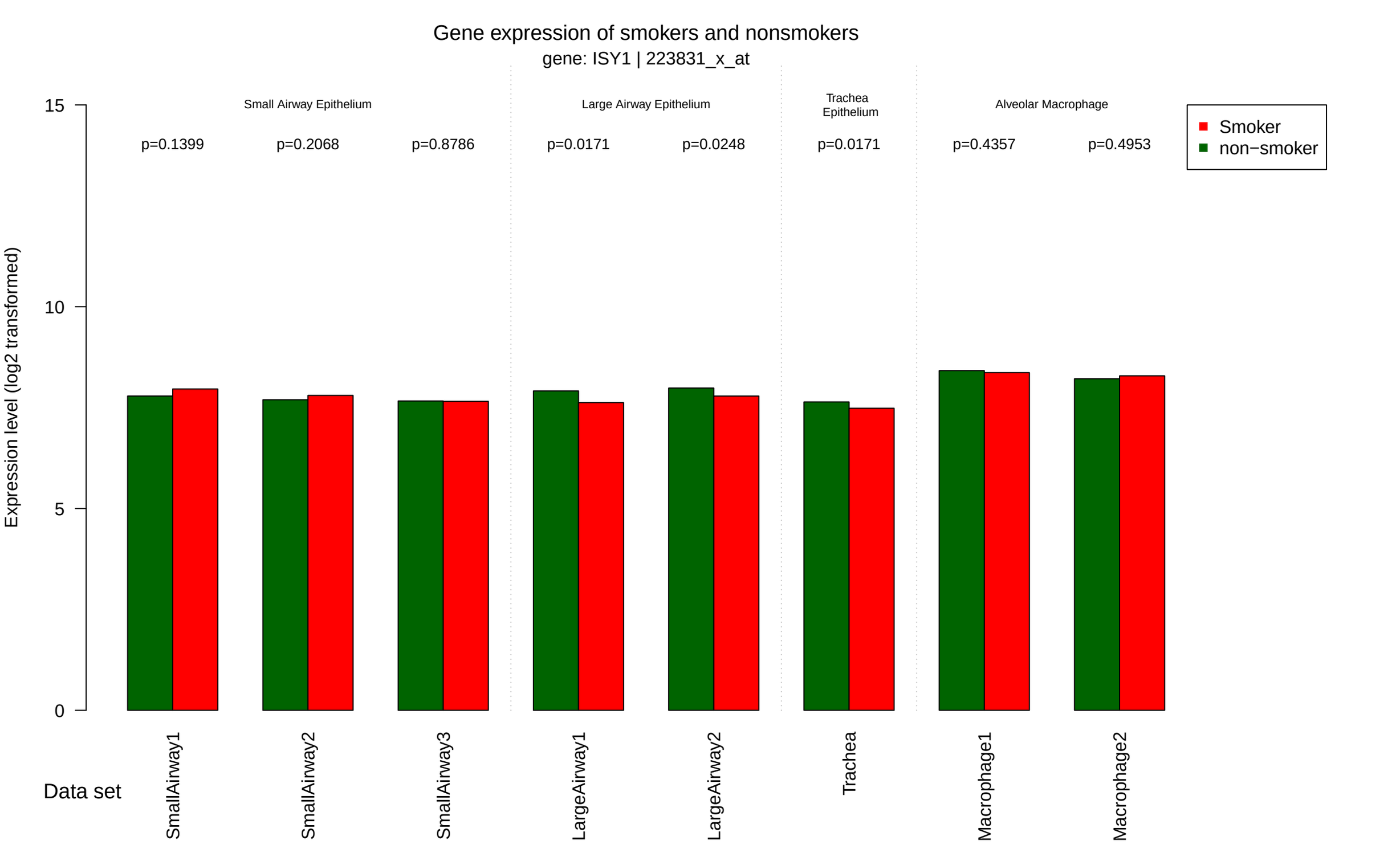
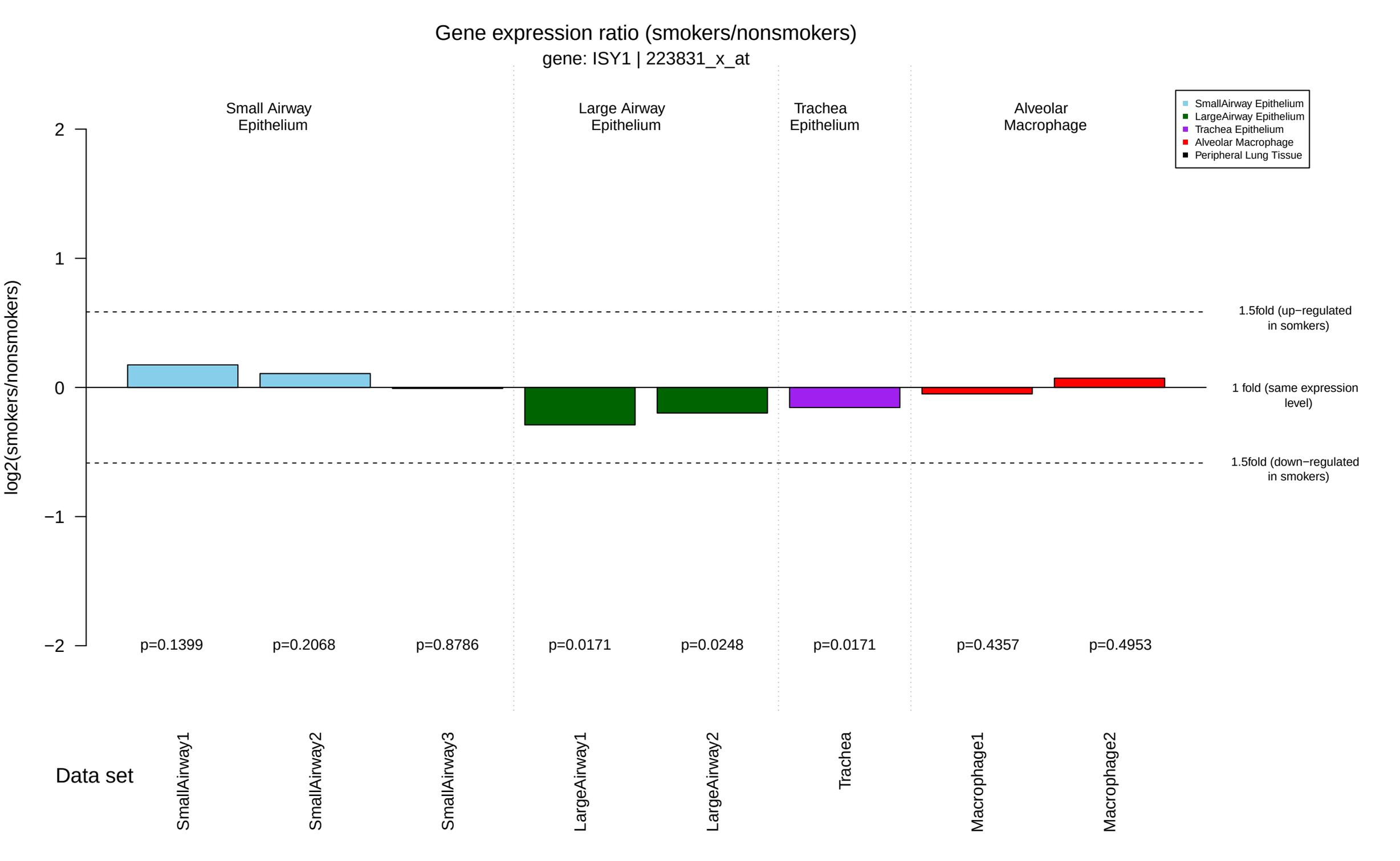
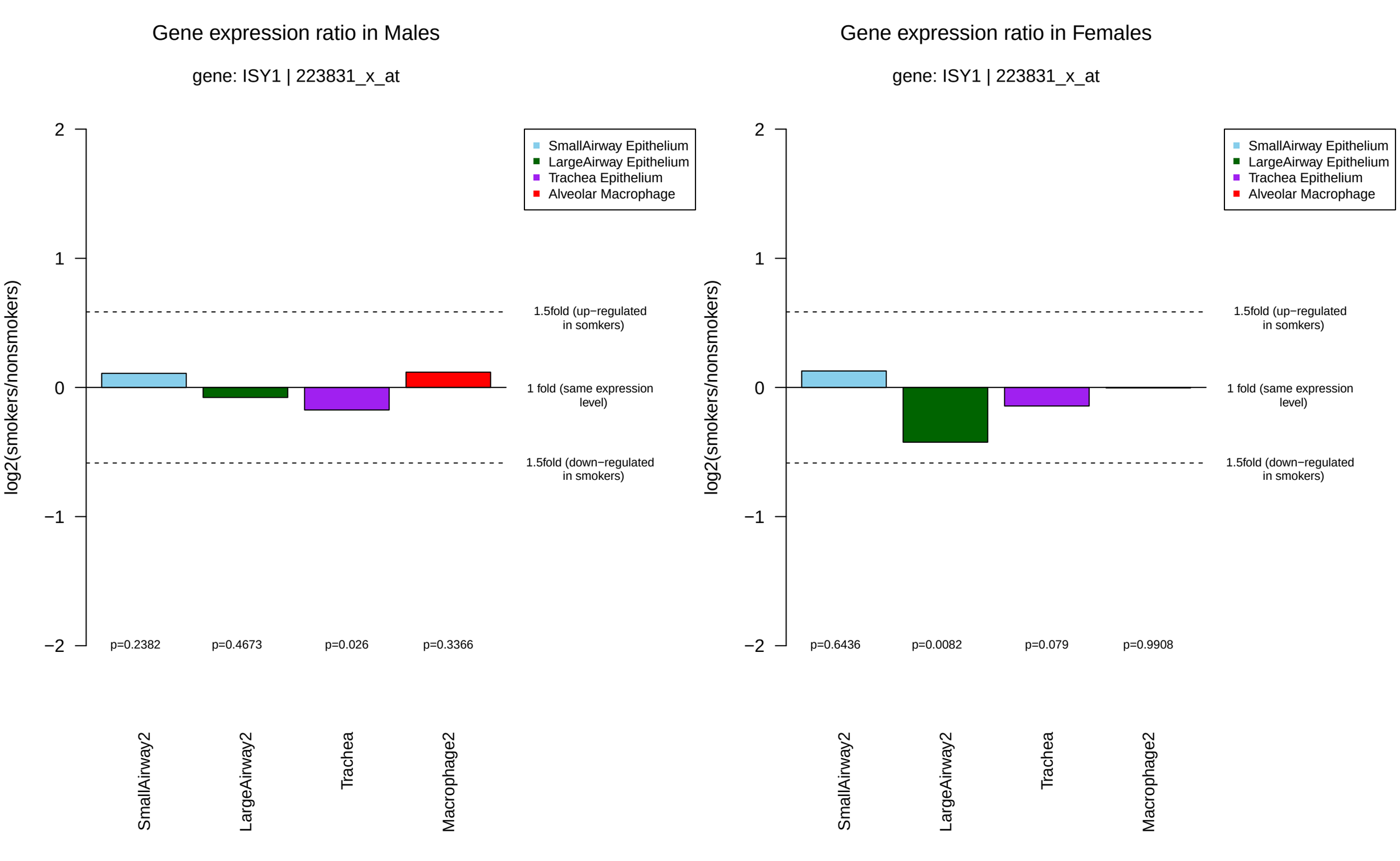
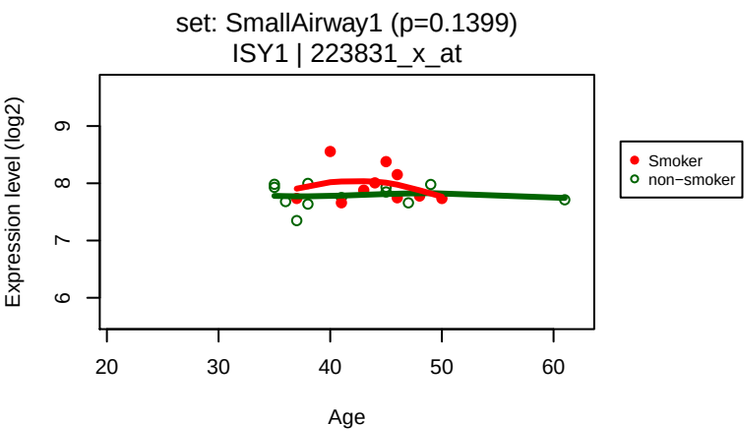
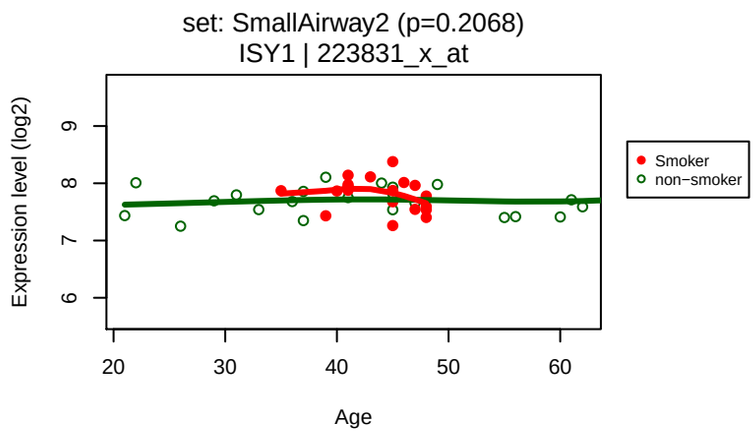
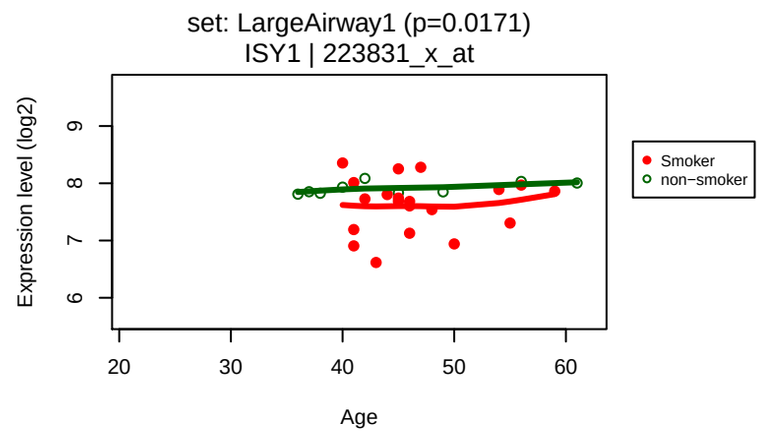

Gene expression of smokers and nonsmokers
gene: ISY1 | 223831_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1399
p=0.2068
p=0.8786
p=0.0171
p=0.0248
p=0.0171
p=0.4357
p=0.4953
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

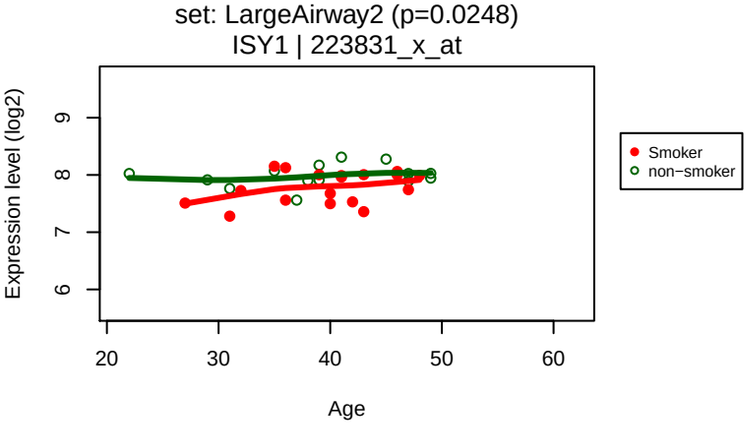

Gene expression ratio (smokers/nonsmokers)
gene: ISY1 | 223831_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1399
p=0.2068
p=0.8786
p=0.0171
p=0.0248
p=0.0171
p=0.4357
p=0.4953
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

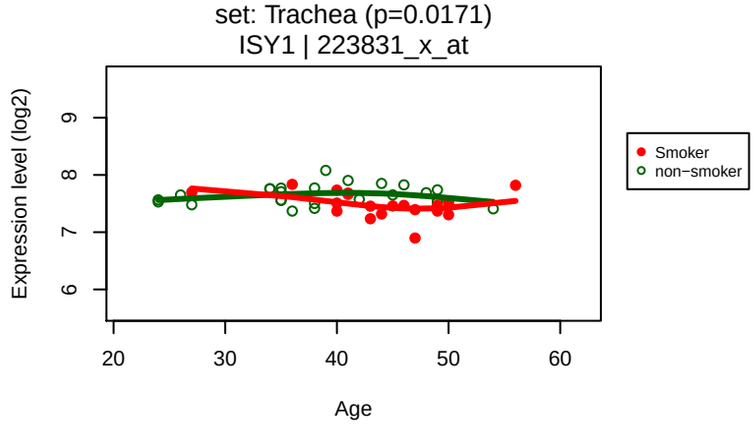

Gene expression ratio in Males
gene: ISY1 | 223831_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2382
p=0.4673
p=0.026
p=0.3366
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ISY1 | 223831_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6436
p=0.0082
p=0.079
p=0.9908
SmallAirway2
LargeAirway2
Trachea
Macrophage2

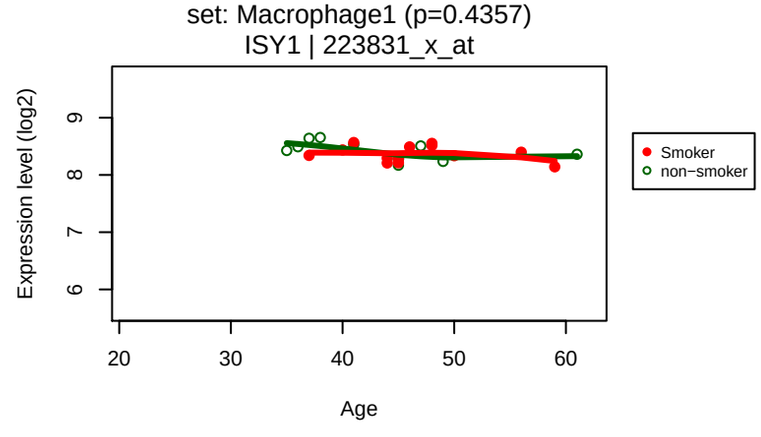

set: SmallAirway1 (p=0.1399)
ISY1 | 223831_x_at
Expression level (log2)
Age
Smoker
non−smoker

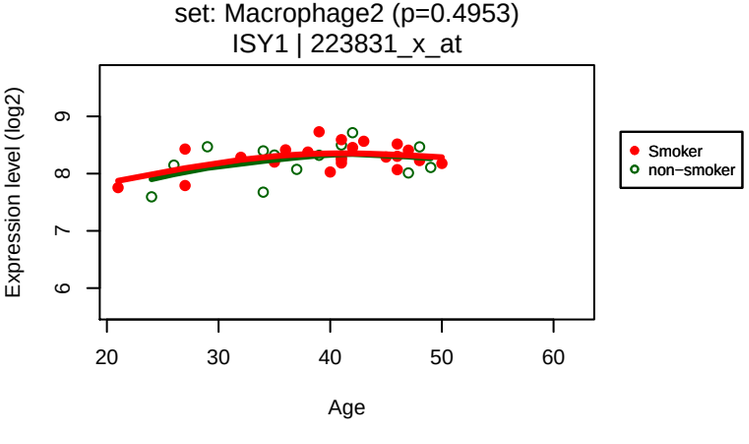

set: SmallAirway2 (p=0.2068)
ISY1 | 223831_x_at
Expression level (log2)
Age
Smoker
non−smoker

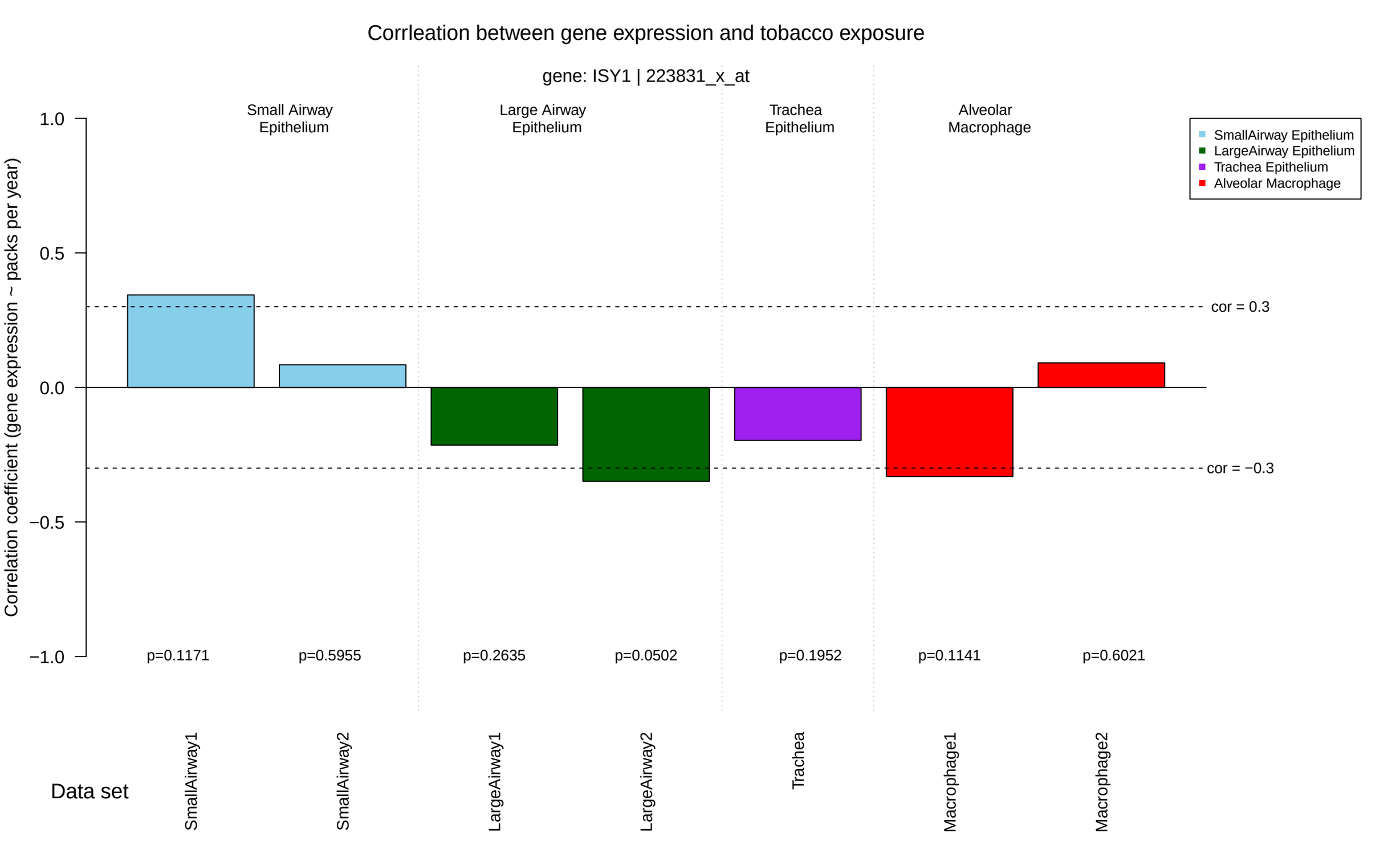

set: LargeAirway1 (p=0.0171)
ISY1 | 223831_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0248)
ISY1 | 223831_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0171)
ISY1 | 223831_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4357)
ISY1 | 223831_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4953)
ISY1 | 223831_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ISY1 | 223831_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1171
p=0.5955
p=0.2635
p=0.0502
p=0.1952
p=0.1141
p=0.6021
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2